Reading Critique: Predictive Representation Learning for World Modeling

1. Problem Statement

Self-supervised learning lacks a principled mechanism for acquiring latent representations that function as *internal state variables* of a dynamical system, rather than as static feature extractors over observations. This perspective is consistent with cognitive theories which argue that intelligent agents form internal models by integrating sensory input in order to predict future states of the world (Craik, 1967; Rao & Ballard, 1999; LeCun et al., 2022). Pixel-level generative modeling approaches optimize likelihood over high-dimensional sensory inputs, a paradigm instantiated by variational autoencoders, autoregressive image models, and pixel-based world models (Kingma et al., 2014; Van Den Oord et al., 2016; Ha & Schmidhuber, 2018; Hafner et al., 2019). Such objectives are mathematically incentivized to allocate capacity to high-entropy, task-irrelevant nuisance variability, conflating perceptual fidelity with the underlying predictive structure of the environment (Vincent et al., 2008; Pathak et al., 2016; Alemi et al., 2016; Burda et al., 2018).

Contrastive representation learning avoids explicit reconstruction but relies on invariances induced by handcrafted data augmentations or negative sampling, yielding representations optimized for discrimination rather than for modeling temporal evolution or action-conditioned dynamics (Oord et al., 2018; Chen et al., 2020; He et al., 2020). While these approaches have demonstrated strong performance on recognition benchmarks, such performance does not imply that a learned embedding constitutes a sufficient summary of past observations for predicting future outcomes (Bengio et al., 2013; Littman & Sutton, 2001). **The core limitation is therefore not the expressiveness of learned features**, but the absence of representations endowed with explicit *predictive semantics*, temporal sufficiency, and uncertainty over future states, which are required for world modeling, belief propagation, and planning (Sutton et al., 1998; Kaelbling et al., 1998).

2. Challenges

Addressing this problem introduces several fundamental challenges. First, learning predictive objectives directly in representation space risks *representation collapse*, where trivial constant encodings satisfy the training objective in the absence of explicit reconstruction losses or contrastive separation (Grill et al., 2020; Tian et al., 2021). Second, retaining pixel-level prediction signals avoids collapse but introduces an opposing difficulty, since likelihood-based objectives necessarily penalize failure to model high-entropy, task-irrelevant observation noise, thereby obscuring the low-dimensional, predictable structure relevant for reasoning and control (Vincent et al., 2008; Burda et al., 2018; Hafner et al., 2019).

Third, **real-world environments exhibit stochastic and multimodal dynamics**, making deterministic latent prediction insufficient for belief updating and planning under uncertainty, yet most self-supervised objectives lack a well-defined probabilistic semantics over future latent states (Sutton et al., 1998; Kaelbling et al., 1998). Finally, *even when temporal prediction is achieved*, it remains unclear when learned representations constitute *predictively sufficient information states*, particularly over long horizons where errors compound and planning complexity grows exponentially (Littman & Sutton, 2001; Destrade et al., 2025).

3. Contributions

3.1. I-JEPA

Prior self-supervised vision methods largely rely on either pixel-level reconstruction objectives or invariance induced through hand-crafted data augmentations, both of which impose inductive biases toward nuisance variability or task-specific assumptions (Bengio et al., 2013; He et al., 2022; Chen et al., 2020). **I-JEPA challenges this paradigm** by positing that semantic structure is more effectively learned through prediction in representation space, operationalized by predicting latent embeddings of withheld image regions rather than their pixel values (Assran et al., 2023). By avoiding direct observation reconstruction, this formulation removes the need to model high-frequency texture and observation noise.

Under this representation-space prediction objective, predicting target-encoder embeddings on ImageNet-1K with linear evaluation and identical ViT backbones achieves 66.9% accuracy, compared to 40.7% when predicting pixels, indicating a substantial gain in semantic quality. This shift concentrates model capacity on predictable, semantically coherent structure while avoiding augmentation-induced biases, consistent with analyses of prediction-based self-supervision (Tschannen et al., 2019; Tian et al., 2021).

I-JEPA further employs a structured masking strategy in which sufficiently large target blocks are predicted from an informative but incomplete context, constraining the learning signal to require object-level or scene-level inference rather than local spatial interpolation (Assran et al., 2023; Pathak et al., 2016). As a result, **the learned representations transfer across tasks with differing abstraction requirements** and can be trained with reduced computational cost due to the absence of a generative decoder, reaching 79.3% linear ImageNet accuracy without view-based augmentations while requiring approximately 2.5× fewer GPU-hours than augmentation-based methods and over 10× fewer than pixel-reconstruction models at comparable scale.

3.2. VJEPA

Deterministic JEPA formulations implicitly assume that minimizing a regression loss is sufficient to characterize future uncertainty, effectively treating prediction errors as homoscedastic and unimodal. This obscures the probabilistic semantics of the learned state and limits applicability in stochastic or partially observable environments (Hafner et al., 2019; Kaelbling et al., 1998). **VJEPA addresses this limitation** by replacing deterministic regression with a variational objective that explicitly models a predictive distribution over future latent states, making uncertainty an intrinsic component of the representation (Huang, 2026).

By adopting a likelihood-free variational formulation, VJEPA preserves the JEPA principle of avoiding observation reconstruction while enabling principled uncertainty propagation and belief updating. The latent representation is reformulated as a predictive information state whose sufficiency for control can be established without pixel likelihoods, aligning the model with predictive state representations and Bayesian filtering (Littman & Sutton, 2001; Kaelbling et al., 1998). **This formulation enables planning over distributions rather than point estimates** and provides formal guarantees against representational collapse through entropy and KL regularization (Tschannen et al., 2019; Locatello et al., 2019).

3.3. V-JEPA 2

Earlier JEPA-based world models implicitly assume that dynamics and control-relevant structure must be learned from interaction data, limiting scalability and generalization due to the scarcity of embodied experience (Ha & Schmidhuber, 2018; Hansen et al., 2023). **V-JEPA 2 relaxes this assumption** by decoupling representation learning from action conditioning, first learning an action-free predictive model from internet-scale video and subsequently grounding it through lightweight action-conditioned post-training (Assran et al., 2025).

This enables large-scale pretraining of predictive representations that support both semantic understanding and long-horizon planning, while avoiding the computational burden and brittleness of video generation. On motion-centric understanding benchmarks, the pretrained V-JEPA 2 encoder achieves 77.3% top-1 accuracy on Something-Something v2 and 82.0% on Kinetics-400, while reaching 39.7 Recall@5 on Epic-Kitchens-100 action anticipation, surpassing prior task-specific models. **These representations are then reused for control** by post-training an action-conditioned predictor, allowing planning to be performed directly in latent space using distance-based objectives and derivative-free optimization (Destrade et al., 2025; Hansen et al., 2023). In closed-loop robot experiments, this enables reliable execution of manipulation skills, achieving 90% grasp success, 100% reach-with-object success, and 50% pick-and-place success on unseen objects in unseen labs, *while receding-horizon execution mitigates compounding prediction errors*.

4. Open Research Directions

For **I-JEPA**, learning exclusively in representation space eliminates observation-level reconstruction but also removes access to image synthesis and likelihood-based evaluation (He et al., 2022; Van Den Oord et al., 2016). More fundamentally, **semantic abstraction is not enforced by the objective itself** but depends critically on the masking strategy (Assran et al., 2023). The sharp degradation in representation quality under suboptimal masking indicates that semantics are induced by carefully engineered heuristics rather than guaranteed by an invariant learning principle, echoing broader findings on the role of inductive biases in unsupervised representation learning (Locatello et al., 2019). This motivates future work on jointly learning or adapting the masking policy during training, so that semantic abstraction emerges robustly across mask distributions rather than being tied to fixed design choices.

VJEPA extends JEPA with a variational formulation that explicitly models uncertainty over future latent states, enabling probabilistic interpretation and formal collapse-avoidance guarantees (Huang, 2026). However, *this added expressivity introduces sensitivity to variational regularization and latent distribution parameterization*, and the model remains likelihood-free with respect to observations. As a result, calibrated sensory prediction and explicit likelihood evaluation remain out of reach, and theoretical sufficiency guarantees rely on information-theoretic assumptions that may not hold uniformly in high-dimensional settings (Bengio et al., 2013). These limitations suggest the need for richer, explicitly multi-modal latent predictors that can represent branching futures while preserving the stability properties of the variational JEPA objective.

In **V-JEPA 2**, scaling self-supervised video pretraining and latent-space planning substantially broadens the applicability of JEPA-based world models, but introduces new alignment challenges (Terver et al., 2025). Planning is performed using distances in representation space whose correspondence to task objectives is not guaranteed *a priori*, and the decoupling of action-free pretraining from action-conditioned post-training can lead to mismatches between learned dynamics and controllable degrees of freedom (Destrade et al., 2025). Moreover, the continued absence of an explicit observation model limits interpretability and complicates failure diagnosis (Ha & Schmidhuber, 2018). ***These issues highlight that while V-JEPA 2 enables practical long-horizon planning without pixel-level rollouts***, it shifts complexity into representation alignment, latent geometry, and control coupling rather than eliminating it.

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